

Lecture 20

12 May 2026

Recap:

Distributions or Generalized Functions

$$\langle \text{distribution}, \text{signal} \rangle = \text{complex number}$$

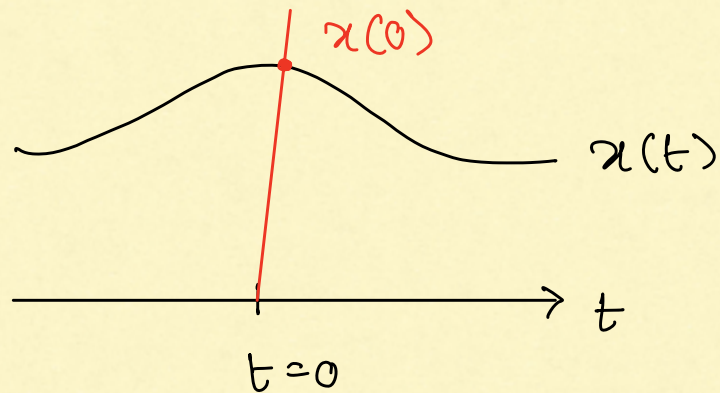
For mathematical consistency, some restrictions are usually placed:

- distributions must act linearly on the signal
- " " " continuously on the signal
- signal must be well-behaved
(differentiability, behaviour as $t \rightarrow \pm\infty$)

Reference: Chap 4 of Osgood's book.

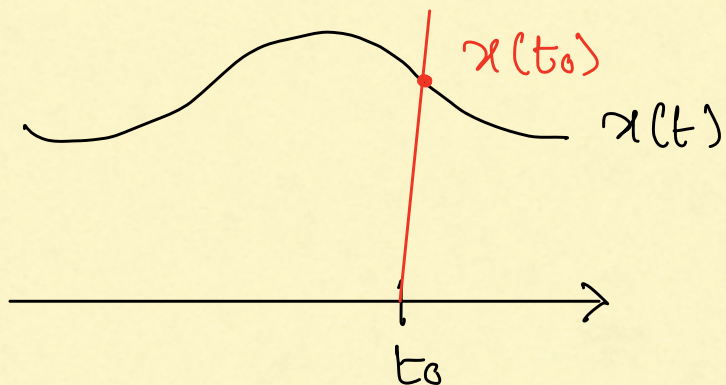
Examples:

① Dirac's Delta δ



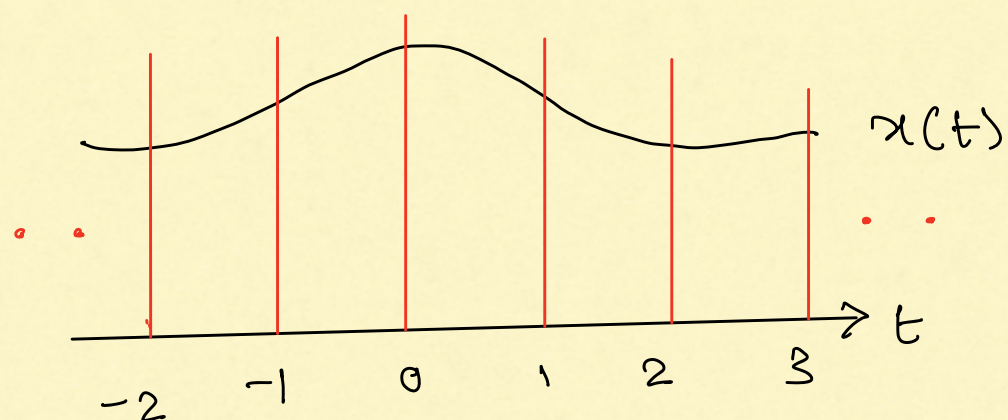
$$\langle \delta, x \rangle = x(0)$$

② Shifted & Scaled Delta



$$\langle a\delta_{t_0}, x \rangle = a x(t_0)$$

③ Shah or Dirac Comb



$$\langle \text{III}, x \rangle = \sum_{k=-\infty}^{+\infty} x(k)$$

Note: For $\langle \text{III}, x \rangle$ to exist $x(t)$ must decay fast enough as $t \rightarrow +\infty$ and $t \rightarrow -\infty$.

(i) If $x(t) = \frac{1}{1+t^2}$ then $\langle \text{III}, x \rangle = \sum_k \frac{1}{1+k^2} < \infty$

(ii) If $x(t) = 1$ or $\frac{1}{1+|t|}$ or e^t , $\langle \text{III}, x \rangle$ is undefined

I

Further Examples

Functions as distributions

A function or a signal $f(t)$ can be used as a distribution

$$\langle f, \chi \rangle = \int_{t=-\infty}^{+\infty} f(t) \cdot \chi(t) dt$$

viewed as a distribution

signal

- We typically assume that χ is sufficiently well-behaved

so that $\int_{-\infty}^{+\infty} f(t) \chi(t) dt$ exists.

Examples

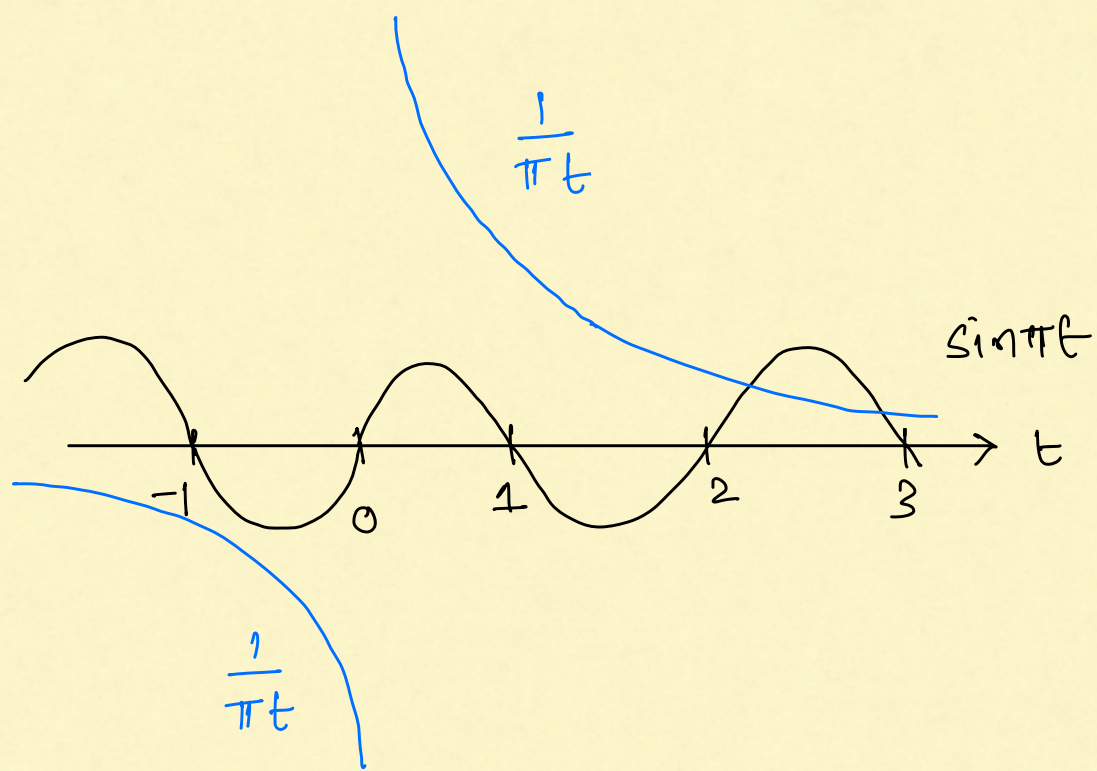
① $u(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$ as a distribution

$$\langle u, x \rangle = \int_0^{\infty} x(t) dt$$

② 1 (the signal that takes value 1 for all t)

$$\langle 1, x \rangle = \int_{-\infty}^{+\infty} x(t) dt$$

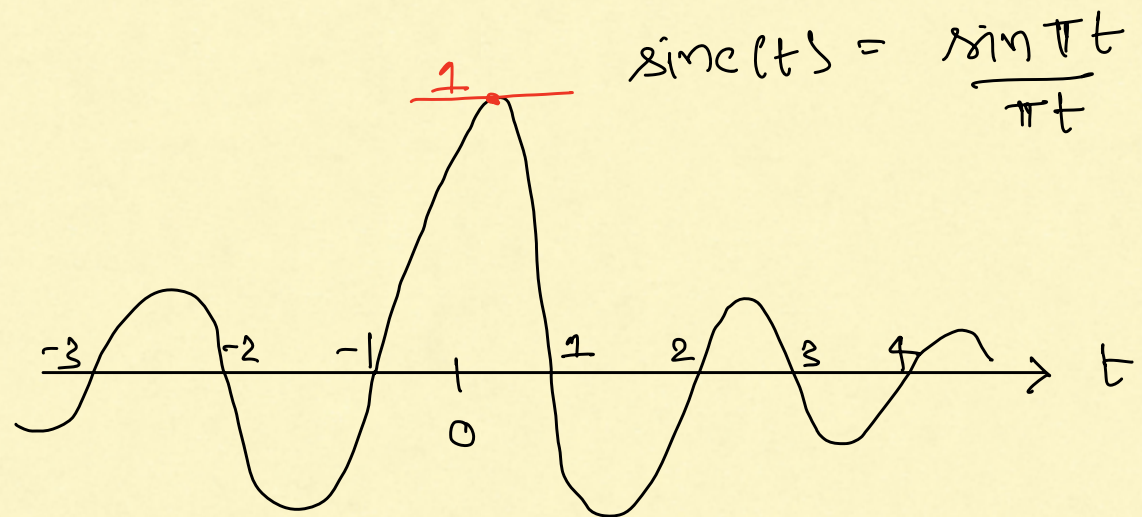
③ $f(t) = \text{sinc}(t) = \frac{\sin \pi t}{\pi t}$



At $t=0$:

$$\frac{\sin \pi t}{\pi t} = 1$$

(This makes the signal continuous at $t=0$)



$$\textcircled{4} \quad f(t) = e^{j\omega_0 t}$$

$$\langle f, x \rangle = \int_{-\infty}^{+\infty} e^{j\omega_0 t} x(t) dt$$

$$\textcircled{5} \quad f(t) = \cos \omega_0 t$$

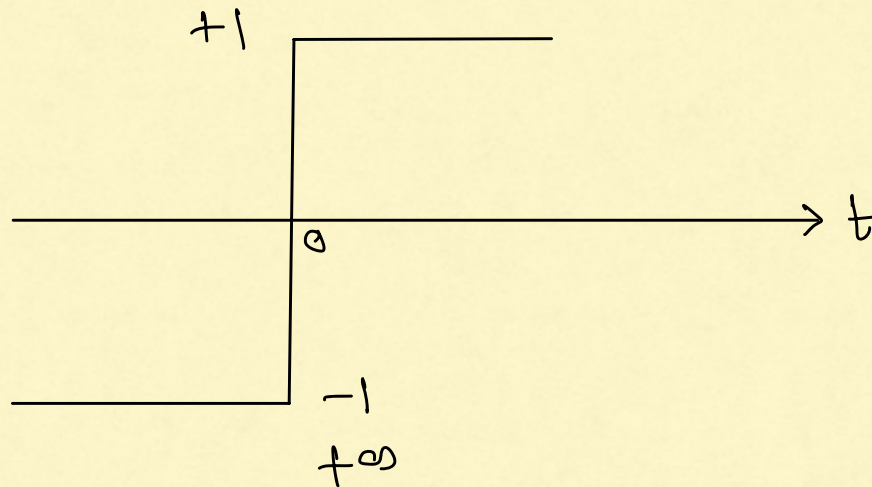
$$\langle f, x \rangle = \int_{-\infty}^{+\infty} x(t) \cos \omega_0 t dt$$

$$\textcircled{6} \quad f(t) = \sin \omega_0 t$$

$$\langle f, x \rangle = \int_{-\infty}^{+\infty} x(t) \sin \omega_0 t dt$$

⑦ Signum Function

$$\text{sgn}(t) = \begin{cases} +1, & t > 0 \\ -1, & t < 0 \\ \frac{1}{2}, & t = 0 \end{cases}$$



$$\begin{aligned} \langle \text{sgn}, x \rangle &= \int_{-\infty}^{+\infty} \text{sgn}(t) x(t) dt \\ &= \int_0^{+\infty} x(t) dt - \int_{-\infty}^0 x(t) dt. \end{aligned}$$

II Distribution as a limit of functions

- This is the traditional viewpoint for analysing distributions.

To a distribution T we associate a sequence of functions:

$$f_1, f_2, f_3, \dots$$

Define $\langle T, \chi \rangle = \lim_{n \rightarrow \infty} \left[\int_{-\infty}^{+\infty} f_n(t) \chi(t) dt \right] \quad \text{--- (1)}$

- We will require χ and $f_1, f_2, \dots$ to be sufficiently well behaved so that the integral and limit in (1) exist.
- Two distinct sequence of functions can describe the same distribution T (if the RHS of (1) is same for all χ).

Example :

Gaussian probability density function (pdf)

$$g(t) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right)$$

$$\text{Mean} = 0$$

$$\text{Variance} = 1$$

For $n = 1, 2, 3, \dots$

$g_n(t)$ = Gaussian pdf with mean = 0, $\text{Var} = \frac{1}{n}$

$$= \frac{1}{\sqrt{2\pi/n}} \exp\left(-\frac{nt^2}{2}\right)$$

- $\int_{-\infty}^{+\infty} g_n(t) dt = 1$

- $g_n(0) = \sqrt{\frac{n}{2\pi}}$

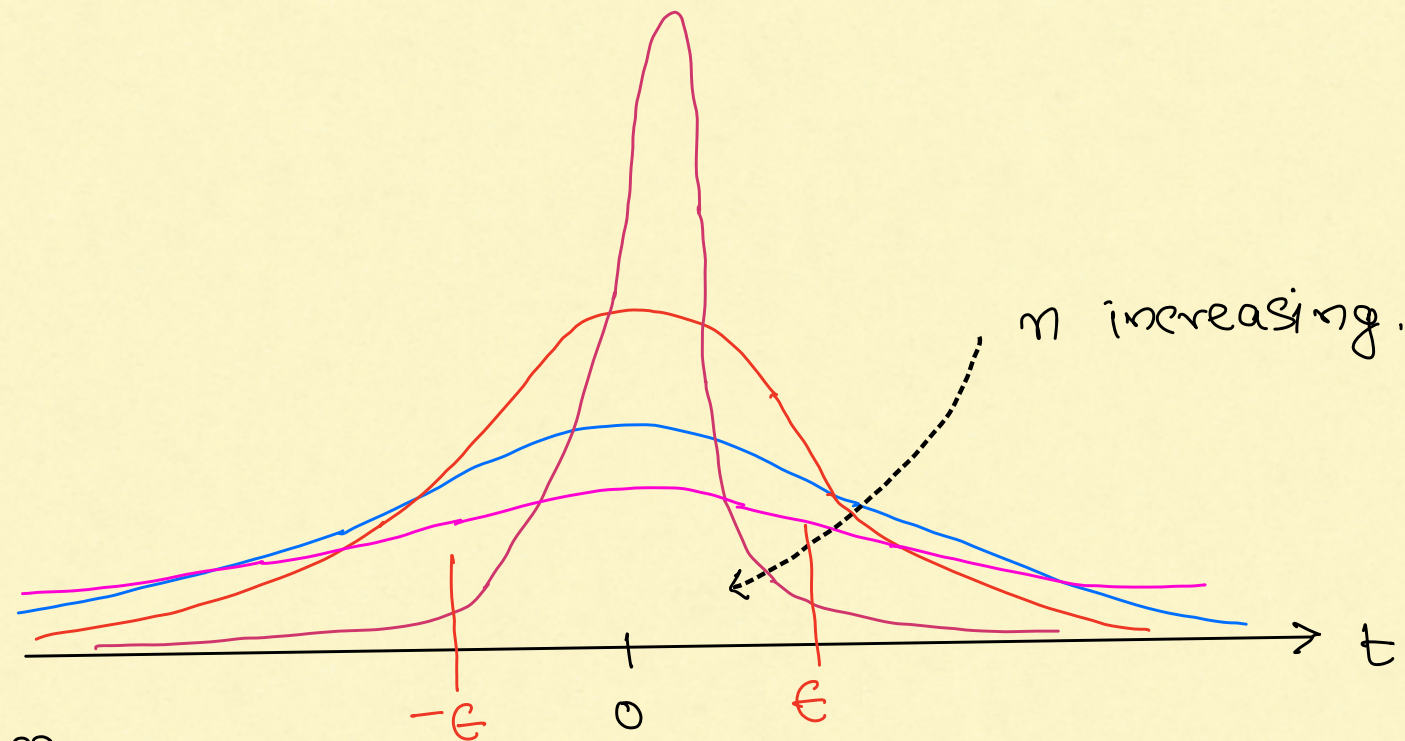
$$\lim_{n \rightarrow \infty} g_n(0) = +\infty$$

- For a given $t \neq 0$:

$$\begin{aligned} \lim_{n \rightarrow \infty} g_n(t) &= \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{\sqrt{2\pi}} (e^{-t^2/2})^n \\ &= 0 \end{aligned}$$

$$g_n(t) =$$

$$\sqrt{\frac{n}{2\pi}} \exp\left(-\frac{nt^2}{2}\right)$$



$$1. \int_{-\infty}^{+\infty} g_n(t) dt = 1$$

$$2. g_n(0) \rightarrow +\infty \quad \text{as } n \rightarrow +\infty$$

$$3. g_n(t) \rightarrow 0 \quad \text{as } n \rightarrow \infty \quad \text{if } t \neq 0.$$

$$4. \epsilon > 0 \Rightarrow \int_{-\epsilon}^{+\epsilon} g_n(t) dt \rightarrow 1$$

Distribution coeffs. to $g_1, g_2, \dots$?

$$\langle T, x \rangle = \lim_{n \rightarrow \infty} \left[\int_{-\infty}^{+\infty} g_n(t) x(t) dt \right]$$

Mathematical precision requires us to assume that $g_1, g_2, \dots$ and $x(t)$ are Schwartz signals^{*}, that is:

- $x, x', x'', \dots$ exist at all t , and
- $|t^m x^{(k)}(t)| \rightarrow 0$ as $t \rightarrow \pm\infty$ for all $m, k \geq 0$

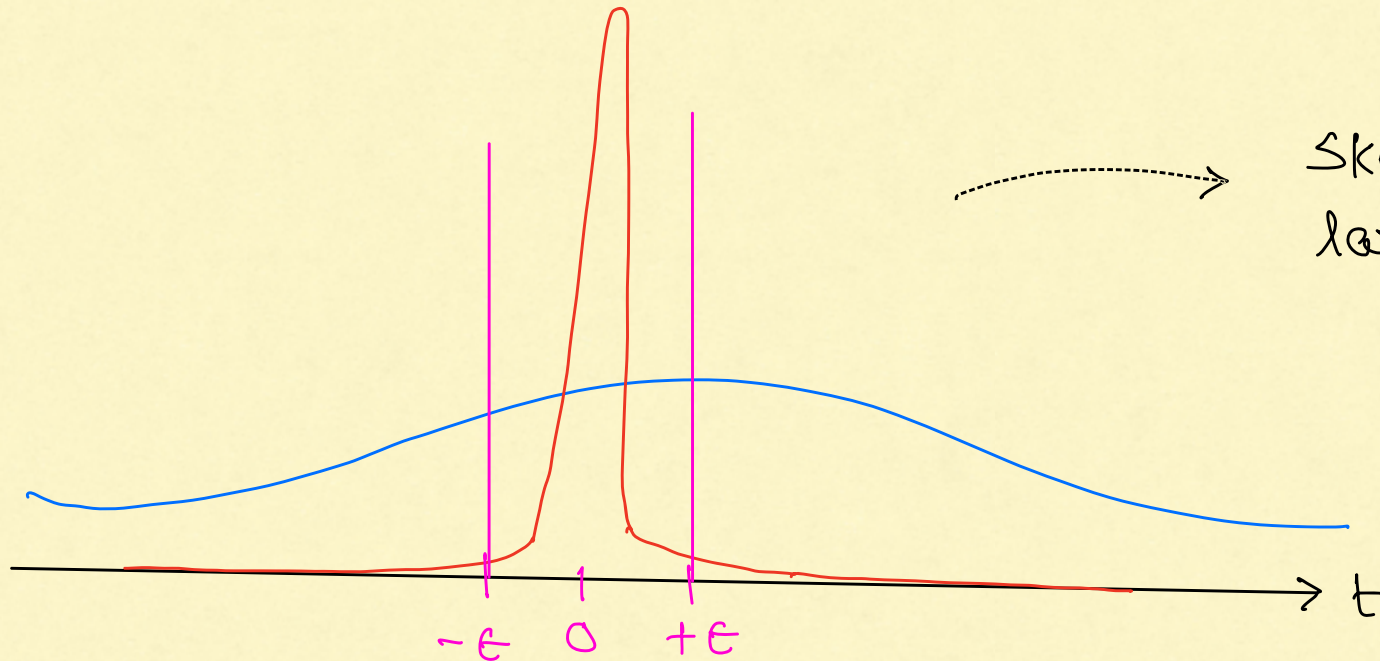
But analysis works for a broader class of signals.

* Ref: Chap 4 of Osgood's book.

Informal Analysis

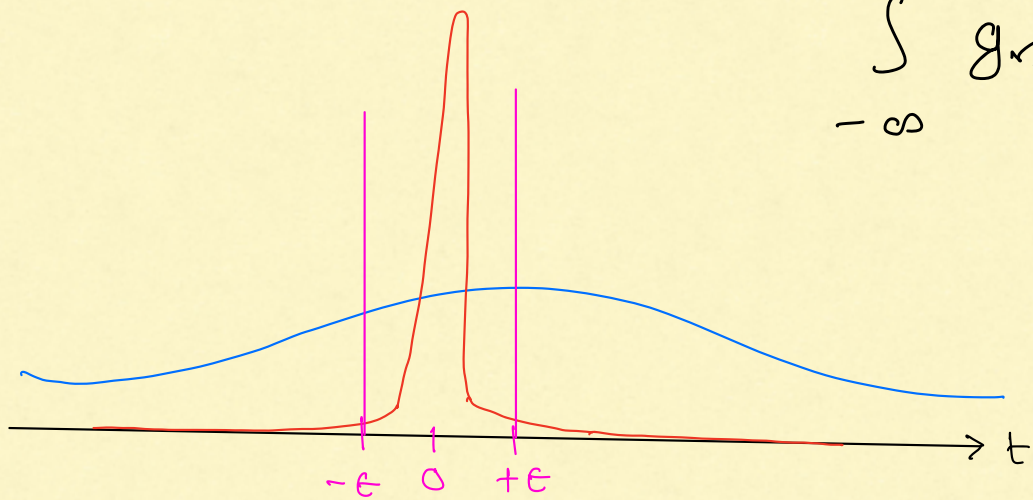
- Pick a small $\epsilon > 0$.

- Consider
$$\int_{-\infty}^{+\infty} g_n(t) x(t) dt$$



Sketch for large value of n .

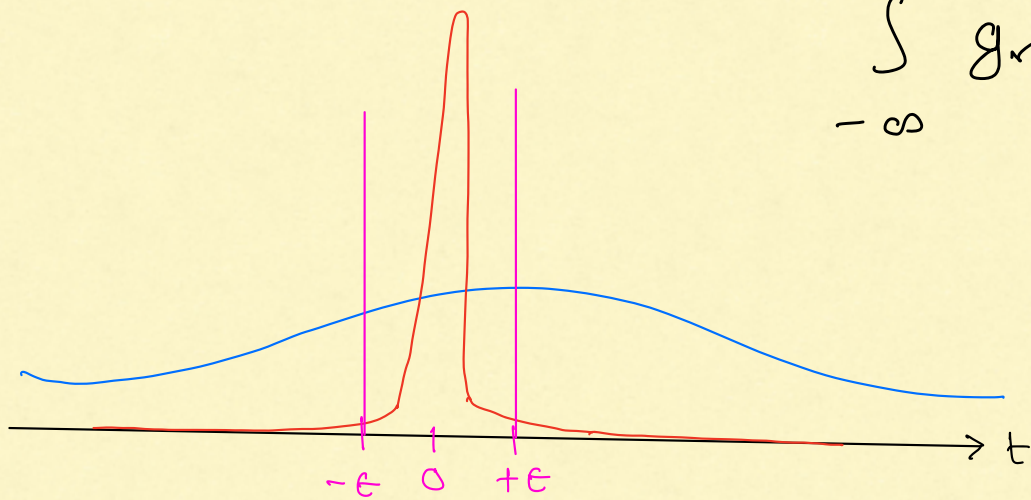
- If $-\epsilon \leq t \leq +\epsilon$ $x(t) \approx x(0)$ if x is continuous at t
- $\int_{-\epsilon}^{+\epsilon} g_n(t) dt \rightarrow 1$ as $n \rightarrow \infty$.



$$\int_{-\infty}^{+\infty} g_n(t) x(t) dt =$$

$$\int_{-\epsilon}^{+\epsilon} g_n(t) x(t) dt \quad \text{--- (a)}$$

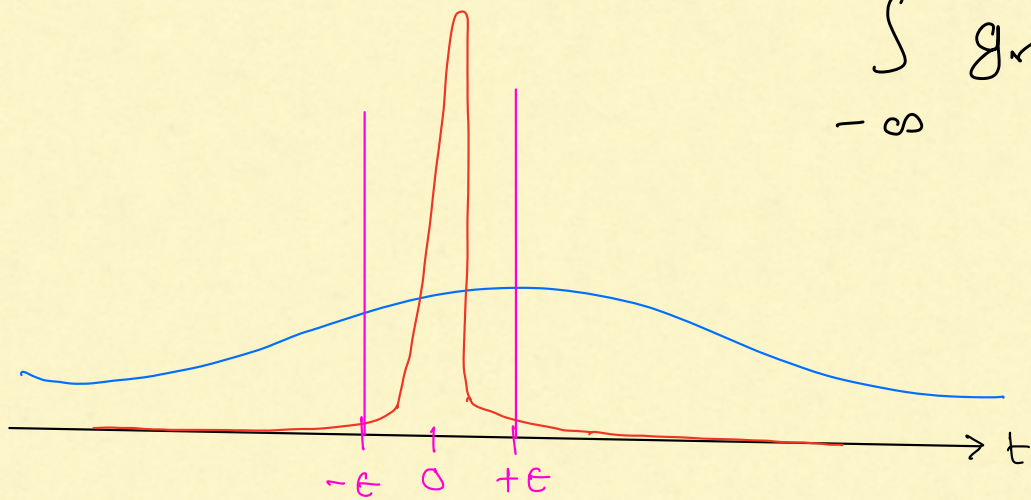
$$+ \int_{|t| > \epsilon} g_n(t) x(t) dt \quad \text{--- (b)}$$



$$\int_{-\infty}^{+\infty} g_n(t) x(t) dt = \int_{-\epsilon}^{+\epsilon} g_n(t) x(t) dt \quad \text{--- (a)}$$

$$+ \int_{|t| > \epsilon} g_n(t) x(t) dt \quad \text{--- (b)}$$

$$\begin{aligned} \textcircled{a} \quad \int_{-\epsilon}^{+\epsilon} g_n(t) x(t) dt &\approx \int_{-\epsilon}^{+\epsilon} g_n(t) x(0) dt \\ &= x(0) \int_{-\epsilon}^{+\epsilon} g_n(t) dt \rightarrow x(0) \times 1 \text{ as } n \rightarrow \infty. \end{aligned}$$



$$\int_{-\infty}^{+\infty} g_n(t) x(t) dt = \int_{-\epsilon}^{+\epsilon} g_n(t) x(t) dt \quad \text{--- (a)}$$

$$+ \int_{|t| > \epsilon} g_n(t) x(t) dt \quad \text{--- (b)}$$

$$\begin{aligned} \text{(a)} \quad \int_{-\epsilon}^{+\epsilon} g_n(t) x(t) dt &\approx \int_{-\epsilon}^{+\epsilon} g_n(t) x(0) dt \\ &= x(0) \int_{-\epsilon}^{+\epsilon} g_n(t) dt \rightarrow x(0) \times 1 \text{ as } n \rightarrow \infty. \end{aligned}$$

$$\text{(b)} \quad \int_{|t| > \epsilon} g_n(t) x(t) dt \rightarrow 0 \quad \text{if } g_n(t) x(t) \text{ is sufficiently small for } t < -\epsilon \text{ \& } t > \epsilon.$$

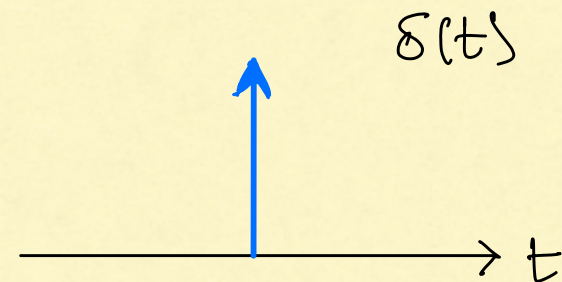
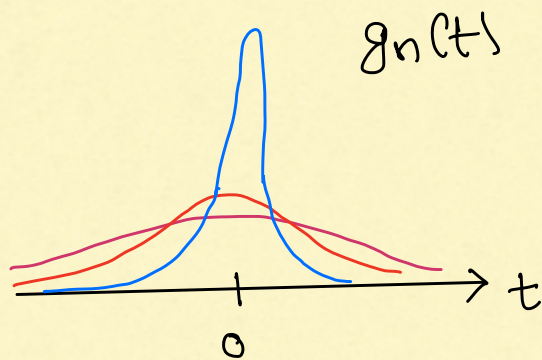
$$\therefore \int_{-\infty}^{+\infty} g_n(t) x(t) dt \rightarrow x(0) \text{ as } n \rightarrow \infty.$$

Theorem: The sequence of functions $g_1, g_2, \dots$

$$g_n(t) = \sqrt{\frac{n}{2\pi}} \exp(-nt^2/2)$$

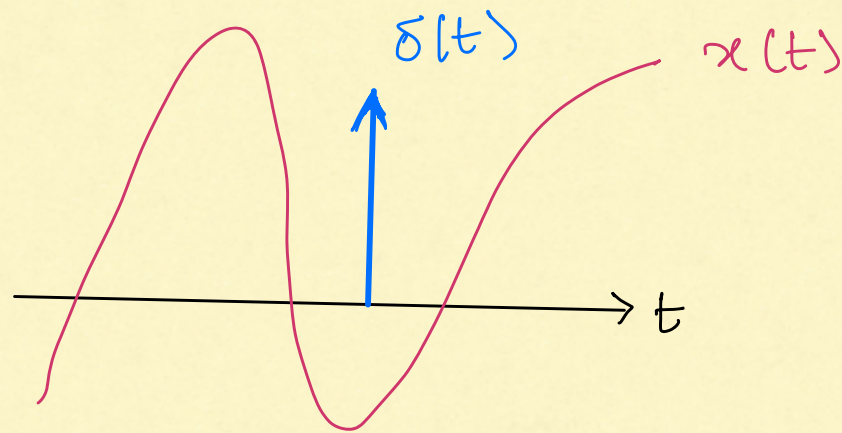
identifies the Dirac's delta distribution δ .

$$\langle \delta, \chi \rangle = \lim_{n \rightarrow +\infty} \left[\int_{-\infty}^{+\infty} g_n(t) \chi(t) dt \right].$$



Graphical Notation for δ
often written as $\delta(t)$.

"Integration" / "Function" Notation



$$\langle \delta, x \rangle = \int_{-\infty}^{+\infty} \delta(t) x(t) dt = \lim_{n \rightarrow \infty} \left[\int_{-\infty}^{+\infty} g_n(t) x(t) dt \right]$$

from
Osgood's book

Standard EE
textbooks

Not an actual integration!

$\delta(t)$ is not an actual function!

"Function" / "Integration" Notation: In general

If T is a distribution associated with $f_1, f_2, \dots$

"Function" Notation: $T(t)$

$$\langle T, \alpha \rangle = \int_{-\infty}^{+\infty} T(t) \alpha(t) dt$$

$$= \lim_{n \rightarrow \infty} \left[\int_{-\infty}^{+\infty} f_n(t) \alpha(t) dt \right]$$

- $T(t)$ is a NOT a function

Note:

It is common to write:

$$\textcircled{1} \int_{-\infty}^{+\infty} \delta(t) dt = 1$$

$$\rightarrow \langle \delta, 1 \rangle = 1$$

$$\textcircled{2} \delta(t) = 0 \quad \text{if } t \neq 0$$

$$\textcircled{3} \delta(0) = +\infty.$$



δ is not a function
"value of δ at time t "
does not make any sense.

$\textcircled{2}$ & $\textcircled{3}$ are not mathematically correct.

Example

Evaluate $\int_{-\infty}^{+\infty} \delta(t) u(t) dt$.

Solution:

First, we recognize the meaning of this integration to be

$$\lim_{n \rightarrow \infty} \left[\int_{-\infty}^{+\infty} g_n(t) u(t) dt \right] \longrightarrow \textcircled{1}$$

since δ is identified by $g_1, g_2, \dots$ where

$$g_n(t) = \sqrt{\frac{n}{2\pi}} \exp\left(-\frac{nt^2}{2}\right).$$

To find the value of the limit in (1), we first do the integration.

Remember: do the integration first and then take the limit.

$$\int_{-\infty}^{+\infty} g_n(t) u(t) dt = \int_0^{\infty} g_n(t) dt$$

= area under one half of the
Gaussian bell curve (mean = 0)

$$= \frac{1}{2}$$

$$\therefore \lim_{n \rightarrow \infty} \left[\int_{-\infty}^{+\infty} g_n(t) u(t) dt \right] = \lim_{n \rightarrow \infty} \left[\frac{1}{2} \right] = \frac{1}{2}$$

Thus,
$$\int_{-\infty}^{+\infty} \delta(t) u(t) dt = \frac{1}{2} .$$